Photosynthesis

Beginner · 45 minutes · version 2

By the end of this lesson you will be able to:

- Describe the inputs and outputs of photosynthesis.
- Explain the role of chlorophyll and sunlight in the photosynthetic process.

Intro and goal

Plants feed the world by transforming simple ingredients into life-giving energy using sunlight.

Example - Imagine a bright green houseplant sitting on a windowsill, soaking up the afternoon sun and growing steadily without ever eating soil or food from a package.

- Describe the inputs and outputs of photosynthesis.
- Explain the role of chlorophyll and sunlight in the photosynthetic process.

- Sunlight as power source
- Leaf as solar factory
- Water and air as raw materials
- Sugars as finished product

Visual: A plant functions like a solar-powered factory.

Overview of Photosynthesis

Photosynthesis is the foundational process that converts light energy into chemical energy.

Organisms like plants and algae capture light energy to fuel their metabolism and produce organic compounds.

Definition - Photosynthesis: A system of biological processes by which autotrophic organisms convert light energy into chemical energy.

Definition - Chlorophyll: A green pigment found in chloroplasts that allows plants to absorb light energy.

Definition - Carbon fixation: The process where carbon dioxide is converted into sugars.

Definition - Photolysis: The splitting of a water molecule by light energy to release oxygen and replace electrons.

- Sunlight → Photosynthesis: drives
- Chlorophyll → Photosynthesis: enables
- Water → Photosynthesis: input
- Carbon Dioxide → Photosynthesis: input
- Photosynthesis → Sugars: output
- Photosynthesis → Oxygen: byproduct

Visual: The core relationships in photosynthesis.

Core idea 1: Inputs and Outputs of Photosynthesis

Photosynthesis takes in water, carbon dioxide, and light energy to produce carbohydrates and oxygen.

Category	Substance	Role
Input	Water	Electron donor and mass source
Input	Carbon Dioxide	Converted into sugars via carbon fixation
Output	Carbohydrates	Stored chemical energy (sugars)
Output	Oxygen	Released as a byproduct

Visual: Summary of photosynthetic inputs and outputs.

Plants build their biomass primarily from carbon dioxide and water rather than soil nutrients alone.

Example - When Jan van Helmont weighed a growing tree and its soil, he found the soil mass changed very little, proving the plant's massive growth came from water and air inputs.

- Water + Carbon Dioxide → Light Energy Capture
- Light Energy Capture → Carbon Fixation
- Carbon Fixation → Carbohydrates + Oxygen

Visual: Flow from raw inputs to final outputs.

1. What gas is released as a byproduct during oxygenic photosynthesis?

Hint: Think about what humans and animals breathe out versus what plants release.

Core idea 2: The Role of Chlorophyll and Sunlight

Sunlight provides the photons that chlorophyll absorbs to initiate electron flow and energy storage.

- Chloroplast structure → Light-harvesting antenna
- Light-harvesting antenna → Chlorophyll pigment
- Chlorophyll pigment → Reaction center
- Reaction center → Electron transport chain

Visual: Path of light absorption inside the chloroplast.

Chlorophyll a absorbs violet-blue and orange-red light while reflecting green, making leaves appear green.

Example - Chlorophyll acts like a solar panel collector: it catches light photons, releases high-energy electrons, and starts the chemical machinery moving.

Light + 6CO2 + 6H2O -> C6H12O6 + 6O2

- Light: Energy source from sunlight
- CO2: Carbon dioxide gas from air
- H2O: Water absorbed from soil
- C6H12O6: Glucose sugar
- O2: Oxygen byproduct

Visual: The net chemical equation for oxygenic photosynthesis.

2. Which green pigment in chloroplasts allows plants to absorb energy from light?

Hint: It is the pigment that gives leaves their green color.

Worked example: Tracing Photosynthesis Step-by-Step

Follow how light, water, and carbon dioxide become oxygen and sugar.

- Step 1: A plant absorbs water from the soil through its roots. Why: Water acts as the essential electron donor.
- Step 2: Sunlight shines on the leaves. Why: Photons provide the necessary energy input.
- Step 3: Chlorophyll pigments absorb the light. Why: This triggers photoinduced charge separation and loosens electrons.
- Step 4: Water molecules undergo photolysis. Why: Splitting water replaces lost electrons and releases oxygen.
- Step 5: Carbon dioxide is fixed into carbohydrates. Why: This stores chemical energy as sugars for the plant.

- Root absorbs water → Leaf captures sunlight
- Leaf captures sunlight → Chlorophyll absorbs photon
- Chlorophyll absorbs photon → Water splits into oxygen
- Water splits into oxygen → Carbon dioxide fixed into sugar

Visual: Step-by-step sequence of photosynthetic events.

Stage	Key Component	Result
Light Reaction	Chlorophyll + Sunlight	Oxygen released & ATP made
Carbon Fixation	Carbon Dioxide	Sugars synthesized

Visual: Overview of light and fixation stages.

Practice: Try it: Mapping Photosynthesis Elements

List the two main inputs and two main outputs of photosynthesis, and explain in one sentence what role chlorophyll plays when sunlight hits the leaf.

Test

You have 10 questions to test your understanding of photosynthesis inputs, outputs, chlorophyll, and sunlight. Hints are available, and answers appear after submission.

Objective	Topic Covered	Question Count
LO-01	Inputs and Outputs	5 Questions
LO-02	Chlorophyll and Sunlight	5 Questions

Visual: Assessment coverage breakdown.

3. What is the ultimate source of the electrons in oxygenic photosynthesis?

- Water
- Carbon dioxide
- Glucose
- Nitrogen

Hint: It is the liquid a plant absorbs from the soil.

4. What are the main organic carbohydrate products of photosynthesis?

- Sugars like glucose
- Mineral salts
- Amino acids
- Fatty acids

Hint: Plants produce sweet organic compounds for energy storage.

5. During carbon fixation, carbon dioxide is converted into what type of compound?

- Carbohydrates
- Water vapor
- Oxygen gas
- Chlorophyll

Hint: Carbon is built into energy-rich sugars.

6. Which pigment in the reaction centre gives up an electron when it absorbs light?

- Chlorophyll a
- Carotene
- Anthocyanin
- Melanin

Hint: It is the specific main type of chlorophyll.

7. Why do plant tissues typically appear green to our eyes?

- Chlorophyll poorly absorbs green wavelengths and reflects them
- Green light is the most useful wavelength for energy conversion
- Plants absorb all green light completely
- Chlorophyll destroys green light

Hint: The color you see is the one that is reflected rather than absorbed.

8. Order the steps of how light energy initiates photosynthesis:

Put these steps in order:

- Chlorophyll absorbs photon
- Electron is loosened
- Electron transport chain starts
- Sunlight photon reaches leaf

Hint: Start with light hitting the leaf and end with electron transport.

9. Match each photosynthetic component to its role:

Match:

- Water = ____
- Carbon dioxide = ____
- Sunlight = ____
- Chlorophyll = ____

Hint: Match each input or catalyst with its direct function.

10. A researcher places a plant in an airtight chamber with water and light, but removes all carbon dioxide. What happens to sugar production?

- Sugar production stops because carbon dioxide is needed for carbon fixation
- Sugar production increases because water is abundant
- Sugar production continues normally using only water
- Sugar production shifts to making proteins instead

Hint: Carbon dioxide is an indispensable input for creating sugars.

11. If a mutation causes a plant's chlorophyll to fail at absorbing violet-blue light, what direct consequence follows?

- The light-dependent reaction efficiency drops because photon absorption is impaired
- The plant stops needing water entirely
- Carbon dioxide absorption increases to compensate
- Oxygen production doubles immediately

Hint: If the solar collector fails, energy capture decreases.

12. Evaluate how removing sunlight would impact both the light reactions and carbon fixation in a plant.

- Both processes eventually halt because light energy powers electron flow and sugar building
- Only carbon fixation stops while light reactions continue indefinitely
- Only light reactions stop while sugar building speeds up
- Neither process is affected because plants store unlimited energy

Hint: Light is the fundamental energy engine for the entire photosynthetic cycle.

Recap

You have mastered the basics of how plants convert light and air into life-sustaining energy.

- Photosynthesis takes in water, carbon dioxide, and sunlight.
- Chlorophyll absorbs light energy and initiates electron transport.
- Water splitting releases oxygen as a vital byproduct.
- Carbon dioxide is fixed into carbohydrates to store energy.

Concept	Key Fact	Primary Role
Water	Absorbed from soil	Electron donor / Oxygen source
Carbon Dioxide	Taken from air	Converted into sugars
Chlorophyll	Green pigment	Absorbs light photons
Sunlight	Light energy	Drives electron flow

Visual: Quick cheat sheet for photosynthesis.

Next steps:

- Observe a green plant outdoors and consider how sunlight fuels its growth.
- Explore how cellular respiration is the reverse process of photosynthesis.

Answer key

1. (Check) oxygen - Water splitting during the light reactions releases oxygen as a byproduct.

2. (Check) chlorophyll - Chlorophyll is the primary green photochemical catalyst that absorbs light energy.

3. (Test) Water - Water is oxidized to supply the electrons during oxygenic photosynthesis.

4. (Test) Sugars like glucose - Photosynthesis stores energy in carbohydrates such as glucose.

5. (Test) Carbohydrates - Carbon fixation converts carbon dioxide into carbohydrates.

6. (Test) Chlorophyll a - Chlorophyll a acts as the pigment in the reaction centre that gives up an electron when it absorbs light.

7. (Test) Chlorophyll poorly absorbs green wavelengths and reflects them - Chlorophyll is a poor absorber of green light, reflecting it and making leaves look green.

8. (Test) Sunlight photon reaches leaf → Chlorophyll absorbs photon → Electron is loosened → Electron transport chain starts - Sunlight hits the leaf, chlorophyll absorbs the photon, releases an electron, and starts the transport chain.

9. (Test) Water = Electron donor; Carbon dioxide = Carbon source for sugars; Sunlight = Energy driver; Chlorophyll = Photon absorber - Water donates electrons, carbon dioxide provides carbon, sunlight drives energy, and chlorophyll absorbs photons.

10. (Test) Sugar production stops because carbon dioxide is needed for carbon fixation - Without carbon dioxide, carbon fixation cannot occur to form carbohydrates.

11. (Test) The light-dependent reaction efficiency drops because photon absorption is impaired - Chlorophyll absorption of light is critical for driving the light reactions.

12. (Test) Both processes eventually halt because light energy powers electron flow and sugar building - Sunlight drives the energy needed for both the light-dependent reactions and subsequent carbon fixation.

Videos and references

Video: Photosynthesis 3D animation - https://www.youtube.com/watch?v=-E4EL4bZbXU
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